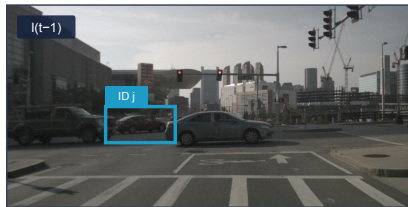


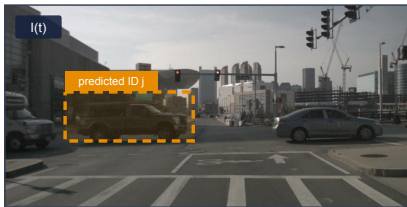
(a) Real nuScenes sequence

sample_003 • conceptual OATM states shown for illustration



Observed

refresh the clear appearance anchor



Hidden

emit a prediction—not a camera detection

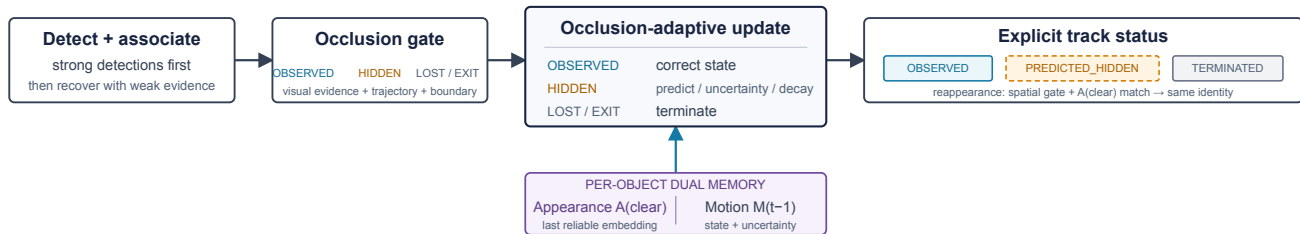


Reappeared

reconnect the original identity

(b) OATM online path

single camera • causal • no future frames



Hidden-state prediction uses camera motion estimated causally from consecutive images.

nuScenes 3D/LiDAR-supported labels are evaluation evidence only; they are not OATM inputs.